

First Terminal Examination – 2080

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

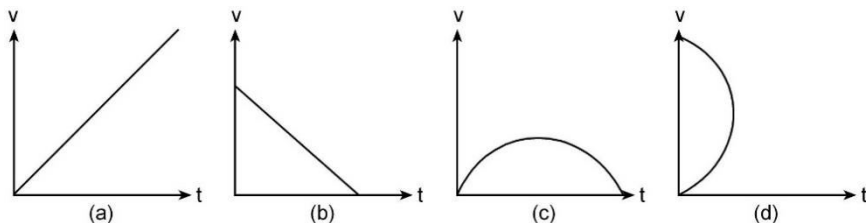
Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. A Student measured the diameter of a wire using a screw gauge with least count of 0.001 cm and listed the measurement. The correct measurement is
 a. 5.3 cm b. 5.32 cm c. 5.320 cm d. 5.3200 cm
2. Hubble's law states that "speed(v) of recession of a galaxy is proportional to its distance (r)" from us, then

$$v = Hr$$
 The dimensions of Hubble's constant is
 a. $[T^{-1}]$ b. $[M^0L^0T]$
 c. $[MLT^4]$ d. $[MLT^{-1}]$
3. Two quantities A and B have different dimensions. Which mathematical operation given below is physically meaningful?
 a. **Fehler!** b. $A+B$ c. $A-B$ d. none
4. If $\vec{P}, \vec{Q} = |\vec{P} \times \vec{Q}|$, then the angle between $\vec{P}$ and $\vec{Q}$ is
 a. **Fehler!** b. 0 c. **Fehler!** d. π

5. Which of the following cannot be speed time graph?



6. Two spheres one solid and other hollow both are made of same material and same radius. If same heat is given to both spheres the expansion will be more in

- a. hollow b. solid c. same d. they do not expand

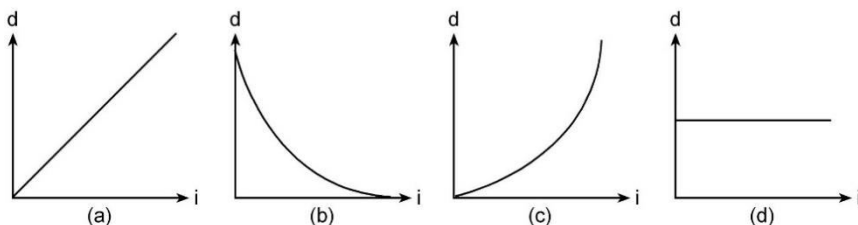
7. When a concave mirror form real image with magnification 'm', the object distance is

- a. $\frac{f}{m}$ b. $\frac{f}{m+1}$
c. $\frac{f}{m-1}$ d. $\frac{f}{m+2}$

8. When light travel from one medium to another, the quantity remain unchanged

- a. Velocity b. wave length c. frequency d. none

9. The variation of lateral shift with angle of incidence is



10. If $\mu_w = 1.33$ and $\mu_d = 2.42$, the refractive index of diamond with respect to water is

- a. 1.8 b. 2.42 c. 0.55 d. 1.3

11. The ray of light entering into a rectangular glass slab emerges from the slab. The incident ray and emergent ray are
- a. Parallel
 - b. Perpendicular
 - c. Makes an acute angle
 - d. Makes an obtuse angle

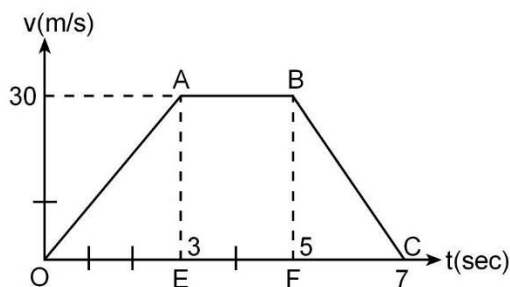
Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a. State the parallelogram law of vector addition. Using this law, find the magnitude and direction of the resultant of two vectors $\vec{P}$ and $\vec{Q}$ inclined at an angle θ from each other. [1+2]
- b. If $\vec{B}$ is added to $\vec{A}$, under what condition does the resultant vector have a magnitude equal to $A + B$? Under what conditions is the resultant vector equal to zero? [2]
13. a. Two forces equal in magnitude have a result equal to either of them in magnitude. At what angle between the two equal forces is this possible? Justify your answer. [2]
- b. A disoriented physics professor drives 3.25 km north, then 4.75 km west and then 1.50 km south. Find the magnitude and direction of the resultant displacement. [3]
14. The velocity time graph is shown in figure.
- a. Find the acceleration of a body from O to A. [1]
 - b. The slope of line AB is zero. What does it mean? [1]

- c. Find the retardation of the body using part BC in the figure. [1]
- d. Draw acceleration time graph using part OA, AB and BC from the figure. [2]



15. a. We are lucky that the earth isn't in thermal equilibrium with the sun which has a surface temperature of about 5800K. But, why aren't they in thermal equilibrium? [2]
- b. A thermometer has wrong calibration. It reads the melting point of ice as -10°C . It reads 60°C in place of 50°C . What is the temperature of boiling point of water on this scale? [3]
16. a. Two metallic rod of the same material but of different length are heated.
Smaller rod has circular area of cross-section but larger rod has rectangular cross-section. [1+1]
- i. Will their linear expansivity be the same or different? Give justification of your answer.
- ii. What about the same material having same dimension. Explain.
- b. A second pendulum made of brass keeps correct time at 10°C . How many second it will lose or gain per day when the temperature of its surrounding rises to 35°C ? [linear expansivity of brass = $18 \times 10^{-6} / ^{\circ}\text{C}$] [3]

17. a. Define mirror formula and prove it for convex mirror.

Fehler! + Fehler! = Fehler!

[1+2]

- b. Find the radius of curvature of a convex mirror, when erect image of **Fehler!** times of object is formed at 6 cm from mirror? [2]
18. a. Derive the expression for apparent shift produced by a medium
Apparent shift (d) = t **Fehler!** [3]
- b. A ray of light incident on glass slab ($\mu = 1.5$) with angle of incident 50° , calculate the lateral shift if the thickness of the glass slab is 8 cm. [2]
19. a. Why does the sun look slightly oval when it is near the horizon? [2]
- b. Calculate the critical angle for (i) an air glass surface (ii) an air water surface (iii) water glass surface. (${}_a\mu_g = 1.5$, ${}_a\mu_w = 1.33$) [3]

Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a. Use velocity- time graph to obtain equation of motion.
i.e $s = ut + \text{Fehler! } at^2$ [3]
- b. A man wishes to swim across a river 600m wide. If he can swim at the rate of 4km/hr in still water and the river flows at 2km/hr.
- i. In what direction must he swim to reach a point exactly opposite to the starting point . [2]
- ii. When will he reach it? [1]
- c. The escape velocity 'v' for the body projected from earths surface of radius 'R' and mass 'M' is written as

$v = \text{Fehler!}$

Check the correctness of the equation using dimensional analysis.

[2]

21. a. A square metal plate is heated from $\theta_1^\circ \text{C}$ to $\theta_2^\circ \text{C}$ so that length

increases from l_1 to l_2 and area from A_1 to A_2 as

$$l_2 = l_1 [1 + \alpha(\theta_2 - \theta_1)]$$

and $A_2 = A_1[1 + \beta(\theta_2 - \theta_1)]$

i. Define the α and β used in above expressions. [2]

ii. Show that $\alpha = \text{Fehler!}$ [2]

b. An aluminium rod when measured with a steel scale, both being at 25°C appears to be one meter long. If the scale is correct at 0°C . [2+2]

i. What is the true length of the rod at 25°C ?

ii. What will be the length of the rod at 0°C ?

Given: [Linear expansivity of aluminium = $26 \times 10^{-6} ^\circ \text{C}^{-1}$,

Linear expansivity of steel = $12 \times 10^{-6} ^\circ \text{C}^{-1}$]

22. a. Define prism and derive an expression for the refractive index of the material of the prism in terms of the refracting angle and the angle of minimum deviation. [1+3]
- b. A small bulb is placed at the bottom of a tank containing water ($\mu = 1.33$) to a depth of 80 cm. What is the area on the surface of water through which light from the bulb emerge out ? [2]
- c. Plane mirror is a special types of spherical mirror. How can you support the statement? [2]

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